

TABLE OF CONTENTS

M-440 EVAPORATIVE DESALINATION PLANT

	Page No.
1. SCOPE OF SUPPLY	M 440-2
2. GENERAL INFORMATION.....	M 440-3
2.1 APPLICABLE STANDARDS AND CODES.....	M 440-3
2.2 OUTLINE.....	M 440-3
2.3 DESIGN DATA.....	M 440-5
2.4 PERFORMANCE GUARANTEE	M 440-6
2.5 TESTING	M 440-6
3. TECHNICAL INFORMATION	M 440-7
3.1 EVAPORATOR AND SEA WATER HEATING SYSTEM.....	M 440-7
3.2 AIR EJECTOR WITH CONDENSER SYSTEM	M 440-9
3.3 PUMPS	M 440-9
3.4 STRAINER	M 440-10
3.5 SCALE INHIBITOR INJECTION FACILITY AND ANTI FOAM INJECTION FACILITY	M 440-10
3.6 DECARBONATOR	M 440-11
3.7 PIPING, VALVES AND SPECIALITIES	M 440-11
3.8 LOCAL CONTROL PANEL	M 440-12
3.9 INSTRUMENTATION.....	M 440-13
3.10 CONTROL SYSTEM	M 440-13
3.11 PLATFORM	M 440-15
3.12 INSULATION	M 440-15
3.13 PAINTING	M 440-15
3.14 CABLES.....	M 440-15
3.15 SPECIAL TOOLS AND STANDARDS TOOL SETS	M 440-15

M 440 EVAPORATIVE DESALINATION PLANT

1. SCOPE OF SUPPLY

- 1.1 The bidder shall provide new Desalination Plant for Block 3 & 4. Each train of Desalination Plant shall be of Flash Evaporation type (Multi Stage Flash type) and shall include, but there shall not be limited to the following equipment and complete accessories:
- a). Steam pressure reducing and desuperheating equipment
 - b). Brine heater
 - c). Flash evaporative and heat recovery stages complete with distillate collection trays and demisting equipment.
 - d). 3 x 50% desalination sea water feed pump for 2 x 100% trains.
 - e). 3 x 50% brine blow down pumps for 2 x 100% trains.
 - f). 3 x 50% distillate delivery pump for 2 x 100% trains.
 - g). 3 x 50% condensate return pump for 2 x 100% trains.
 - h). Two stage air ejection equipment complete with vent condensers
 - i). Chemical solution tank and equipment
 - j). Scale inhibitor tank and equipment
 - k). Chemical storage and dosing equipment
 - l). Piping and valves, foundation bolts and nuts, platform, insulation, painting, cables, and special tools and standard tools.
 - m). All necessary electrical equipment, instrumentation and controls associated with the above equipment for the safety operation of the plant from a panel located in the BOP control building and water treatment building and CRT operator console in BOP control building.
 - n). Scale inhibitor injection facility, drainage facility and anti foam injection facility
 - o). Ball clearing equipment and its associated facility
 - p). Control panel

- q). All chemicals, consumables including spare modules for two year commercial operation.

Note: The Bidder shall provide any and all equipment, materials and accessories as required for a complete set of the system as well as the cost of installation and trial operation of the equipment.

- 1.2 The existing Desalination Plant shall be connected for Block 2.

2. GENERAL INFORMATION

2.1 APPLICABLE STANDARDS AND CODES

The desalination plant shall be designed and constructed in accordance with the Technical General Condition of "Applicable Standards and Codes".

2.2 OUTLINE

- (1) The desalination plant shall be selected under the following conditions.
 - (a) The system shall be of proven design, and has a good reputation and long experience in steam power stations. The experience record shall be submitted.
 - (b) One that can be operated and maintained easily particularly shall use minimum number of pumps.
- (2) The Bidder shall clearly state in the "Bidder's Data Sheet" the required amounts of electric power, steam consumption and chemicals at the rated load of 1000 Tons/day.
- (3) Installation and trial operation of the desalination plants shall be completed before the date when the hydrostatic test of the first HRSG is carried out, and the plants shall be ready for operation at that time.
- (4) The sea water supply piping for the desalination plants shall be injected with hypochlorite on the internal surface of pipe to prevent the growth of marine organisms and plants.

The injection amount shall be maximum of 0.2 ppm residual chlorine at the desalination plant inlet.
- (5) The desalination plant shall be used to produce the necessary plant makeup water, trial operation water for the unit, etc.

- (6) Two (2) sets of desalination plant shall be installed for Blocks 3 and 4. Two (2) sets of desalination plant shall be operated in parallel and be connected to two (2) new 1500 ton Raw Water Tanks.

Block-2 raw water shall be supplied from the existing Block-1 facilities.

- (7) The desalination plants shall be operated at the rated load, but shall be able to be operated effectively at partial load as well.
- (8) Water for the desalination plants shall be obtained from the desalination seawater feed pumps installed in desalination pump intake pit.
- (9) The distillate water shall be pumped to the raw water tanks.

Brine from the plant shall be discharged to the sea through the brine discharge piping to be connected to main drainage piping at road.

- (10) The steam for the seawater heating system and steam jet air ejector shall be supplied from the low-pressure auxiliary steam line or the auxiliary boiler.
- (11) The heating steam condensate from the desalination plant shall be returned to the main surface condenser. It shall be connected to valve flange provided on condenser hotwell.
- (12) Scale inhibitor injection system shall be installed to prevent rust and deterioration of the heating surface of the plant.
- Anti foam injection equipment shall be provided to prevent of bubbling and foaming in the evaporator chamber.
- (13) All equipment and the control panel for local control shall be located in desalination control building.
- (14) The desalination plants shall be automatically shut down in accordance with the high water level in the raw water tanks.
- (15) The quality of distillate water shall be automatically checked and recorded.

The above instruments and recorder to check the operating conditions of the desalination plant shall be provided on the local control panel.

- (16) All necessary devices for automatic operation such as indicators, recorders, controllers, alarms, etc, shall be provided on the control panel.
- (17) The scale inhibitor cleaning devices for regular cleaning shall be provided by the Bidder.
- (18) The necessary platforms and ladders shall be provided for operation and maintenance.

2.3 DESIGN DATA

- | | | |
|------|---|--|
| (1) | Installation | Outdoors |
| (2) | Type Flash | evaporator type (MSF type) |
| (3) | Number | Two (2) sets |
| (4) | Capacity | 1000 tons/day |
| (5) | Number of stage | Thirteen (13) stages |
| (6) | Performance ratio | 6 or more |
| (7) | Quality of distillate water | |
| | Total dissolved solid | Less than 10 ppm. |
| | Fe | Less than 0.05 ppm. |
| | pH | 6.5 - 7.0 |
| (8) | Temperature of distillate water | Less than 40°C |
| (9) | Quality of sea water | Refer to Table 1.1-1 Seawater Analysis specified IN Clause 1.1.4.7.2 in Section 1 of Part 4. The Bidder is to satisfy himself of the true conditions by means of his own survey, especially for sludge conditions. |
| (10) | Temperature of sea water | Design 30 °C |
| (11) | Steam source for brine heater and air ejector | |
| (a) | Low pressure auxiliary steam | 7 kg/cm ² g [0.69Mpa], 350°C |
| (b) | Auxiliary | 7 kg/cm ² g [0.69Mpa], 183°C |

2.4 PERFORMANCE GUARANTEE

- (1) The desalination plants shall be guaranteed to produce distillate water of the quality and capacity as specified in the design data.

The quality shall be determined by sampling and analysis of produced distillate water every 2 hours during 24 hours continuous operation. The average of quality above sampling and analysis shall be accordance with the design data.

- (2) The desalination plant shall be guaranteed for the items specified in the "Bidder's Data Sheet".

2.5 TESTING

- (1) Shop test

The Bidder shall execute the following shop test and shall submit the test reports to the Employer and Engineer for approval.

- (a) Heat exchangers

- Material test
- Dimension test
- Hydrostatic test
- Non-destructive test

- (b) Pumps

- Material test
- Dimension check and visible inspection
- Hydrostatic test
- Performance test
- Vibration check
- Noise check
- Overhaul inspection

(2) Field test

The Bidder shall execute the following field tests in the presence of the Employer and Engineer and shall submit the test reports to the Employer and Engineer for approval.

- (a) Alignment check
- (b) Water fills test and hydrostatic test
- (c) Running test of pumps and motors
- (d) Performance test

3. TECHNICAL INFORMATION

3.1 EVAPORATOR AND SEA WATER HEATING SYSTEM

- (1) The design pressure of the shell and tube of evaporator, seawater heating system and condenser shall be decided by the Bidder.
- (2) The materials stated in the following are the minimum requirements. If alternate materials are used, the Bidder shall propose equivalent or higher-grade materials.

(a) Evaporator

1. Shell

- Bottom Side : 316LSS + C. Steel Clad
- Stop Plate : 304SS + 316LSS C. Steel
- Distillate Tray and trough : 316LSS

2. Tube

- All Stages : Titanium (t = 0.5 mm)

3. Tube Sheet : Al - Bronze

4. Water Box : Carbon Steel and 90/10
Cu-Ni lining.

(b) Brine Heater

1. Shell : Carbon Steel

2. Tube : Titanium (t = 0.5 mm)

- 3. Tube Sheet : Al - Bronze
- 4. Water Box : Carbon Steel and 90/10
Cu-Ni lining.

(c) Ejector and Ejector Condenser

- 1. Ejector : 316SS
 - Tube : Titanium (t = 0.5 m)
 - Tube Sheet : Al – Bronze
 - Shell : 316LSS
 - Water box : Carbon Steel and 90110
Cu-Ni lining.

(d) Mist Separator : Stainless Steel

(e) All components not listed shall be corrosion proof materials.

- (3) The access manhole shall be provided for maintenance and inspection of the internal part of the evaporator.
- (4) The evaporator condenser shall be provided with an air-cooling zone, and shall be designed to collect non-condensable gases and lead them to the air ejector.
- (5) The vessel shall be equipped with glass gauges for indication of the brine level in each stage and for the evaporator condenser distilled water level.
- (6) All heat exchanger tube shall be of straight tube type.
- (7) The heat exchanger shall be designed in accordance with the specifications of "Heat Exchanger Institute".
- (8) The drain cooler shall be furnished at the outlet of heating steam condensate pump.

The cooling water shall be branched from the main seawater feed line. Outlet steam condensate temperature of the cooler shall be less than 50°C.

3.2 AIR EJECTOR WITH CONDENSER SYSTEM

- 1). Air ejectors shall be provided to create the initial vacuum required for start up of the desalination plants.

During the normal operation, the ejectors shall be used to maintain the vacuum in the evaporator condenser.

- 2). The condenser shall be provided to condense the steam used in the ejectors.
- 3). All components of this system shall be constructed of corrosive-proof alloys.

3.3 PUMPS

- 1). The necessary pumps for the desalination plant shall consist of desalination seawater feed pumps, distillate delivery pumps, and condensate return pumps. The above pumps shall be motor driven. If required brine blows down pumps or brine circulating pumps shall be provided.
- 2). The above pumps shall be of 3x50% capacity for two (2) trains. Which is one (1) set of those pumps shall be as a standby unit for both trains in normal operation
- 3). The pumps shall be provided complete with motors, base plates, couplings, coupling guards and isolating valves.
- 4). Pump design shall be accordance with manufacturer's standards, but the materials shall be of anti-corrosive ones.
- 5). The desalination sea water feed pump shall be vertical type with rubber lining, and shall be of self water lubricating type.

Lubricating water shall be supplied through the double strainers.

- 6). Distillate delivery pumps, condensate return pumps, brine blow down pumps and brine-circulating pumps shall be of the horizontal type.
- 7). The motor shall be in conformity with E211 of "Medium Voltage Motors", E210 of "Low Voltage Motors" in Book 2D.

3.4 STRAINER

- (1) Auto cleaning type
 - (a) Quantity Two (2) sets for two desalination plants
 - (b) Screen 40 mesh
 - (c) Material SS carbon steel inside rubber lining
 - Basket SUS 316L or equivalent
 - Screen SUS 316L or equivalent
- (2) Duplex Type
 - (a) Quantity Four (4) sets for two desalination plants
 - (b) Screen 40 mesh
 - (c) Material SS carbon steel inside rubber lining
 - Basket SUS 316L or equivalent
 - Screen SUS 316L or equivalent
- (3) The strainer shall be made of anti-corrosion material.
- (4) The differential pressure alarm signal of the strainers shall be transmitted to the local control panel of desalination plant.
- (5) Two types of seawater strainers for the desalination plant shall be furnished. The auto cleaning type strainers shall be installed on the desalination seawater feed pump discharge, and the duplex type shall be installed nearby the desalination plant.

3.5 SCALE INHIBITOR INJECTION FACILITY AND ANTI FOAM INJECTION FACILITY

- (1) The Scale inhibitor injection facility shall be provided for preventing the adherence of scale on the surface of condenser tube.
- (2) One (1) solution tank of more than seven (7) days' capacity for scale inhibitor injection facility shall be provided.

The material of solution tank shall be corrosive-proof material.

- (3) Two (2) sets of scale inhibitor injection pump of 100% capacity for each plant shall be installed.

The pump discharge shall be of the automatic control type.

The liquid side of each pump shall be constructed of corrosive-proof material.

- (4) The piping, valves, etc., from solution tank to injection point shall be provided.
- (5) The scale inhibitor transfer pump for the scale inhibitor storage tank shall be provided.
- (6) The drainage piping from scale inhibitor tank area shall be led to the desalination drainage pit, and the drainage water shall be sent to the wastewater pond by the sump pump.
- (7) The drainage piping shall be of corrosion-proof materials.
- (8) One (1) anti foam solution tank of more than seven (7) days capacity for anti foam injection facility shall be provided.
- (9) Two (2) sets of anti foam injection pump of 100% capacity for each plant shall be installed.

The pump discharge shall be of the automatic control type.

The liquid side of each pump shall be constructed of corrosive-proof material.

3.6 DECARBONATOR

The decarbonator with exhaust fan shall be provided to remove carbon dioxide gas.

3.7 PIPING, VALVES AND SPECIALITIES

- (1) All necessary piping, valves, hangers, supports, structures and supporting structures for the desalination plant shall be provided.
- (2) Piping and accessories between desalination seawater feed pumps and desalination plant shall be provided.

- (3) Two types of seawater strainers for the seawater feed pipe of desalination plant shall be furnished. One shall be a simplex type and the other type shall be a duplex type.

The simplex type strainers shall be installed on the desalination seawater feed pump discharge, and the duplex type shall be installed nearby the desalination plant.

- (4) Steam piping for the air ejector and the heating equipment shall be led from low-pressure auxiliary steam line and auxiliary boiler.
- (5) Blow piping of desalination plant, such as brine water piping, etc., shall be led to the sea through the yard drainage pipe.
- (6) Condensate piping from the heating equipment to the main surface condenser and to the feed water tank of auxiliary boiler shall be provided.
- (7) Distillate water piping and accessories shall be provided and led to the raw water tanks.
- (8) All other necessary piping, valves, etc., for the desalination plant shall be provided.
- (9) The piping shall, as a rule, be installed on ground, but blow piping of desalination plant shall be buried.

3.8 LOCAL CONTROL PANEL

- (1) One (1) local control panel shall be installed for the two (2) plants with separate controls and instruments except for cases of common use.
- (2) Remote start/stop control and supervision of the desalination plants shall be possible on the plant DCS.
- (3) The summarized alarm and operating signal shall be transmitted to the CRT operator console in the main control room.
- (4) The instruments for safely operation of the desalination plant shall be provided on the local control panel, but these shall not be limited to the following.

- (a) Flow indicator, recorder and integrating meter
 - (b) pH controller and recorder
 - (c) Conductivity monitor and recorder
 - (d) Temperature controller
 - (e) Vacuum indicator
 - (f) Control switch, indicating lamp, alarm window, etc.
- (5) A control shall contain switch, status lights and the alarm enunciators. These enunciators shall be provided to CRT of local control panel.
- (6) The local control panel shall be of indoor self-standing type.

3.9 INSTRUMENTATION

- (1) All control devices shall be furnished for the desalination plants to enable complete and satisfactory installation.
- (2) The devices to be furnished, shall include all control switches, level switches, solenoid valves, time relays, relays, panels and all other necessary pertinent.
- (3) All controls, instruments and accessories to be mounted locally on the desalination plants shall be of waterproof construction suitable for tropical climate.
- (4) All required piping, tubing, and wiring for instrumentation, including fittings, supports, etc., should be provided.
- (5) All control devices shall be in conformity with I-001 of "Instrument and Control Apparatus".
- (6) The interlock sequence for automatic control of the desalination plant shall be functional under the consideration of tank level, difference between the raw water tanks.

3.10 CONTROL SYSTEM

- (1) Start up and shutdown of operation shall be made possible by one touch system on the control panel and CRT operator console in central control building.

- (2) The plant shall be manually started upon low-level signal of the raw water tanks.
- (3) The rate change of produced water shall be controlled by control valve at the inlet seawater or brine recirculation water amount to the evaporator.
- (4) The plant shall stop automatically upon high water level signal of the raw water tanks.
- (5) The conductivity monitor and recorder shall be furnished on the control panel to check the water quality at the delivery of desalination plant.
- (6) Blow-down of high conductivity distillate water shall be automatically done and alarmed. When the conductivity is recovered, the distillate water shall be supplied to the Raw Water Tank automatically.
- (7) The brine heater outlet temperature control system shall be provided to maintain a constant brine temperature in 1st stage evaporator by throttling the inlet steam to the desalination plant heating equipment.
- (8) The brine level control system shall maintain constant distillate level in the desalination plant.
- (9) Distillate level control system shall maintain constant distillate level in the desalination plant.
- (10) The condensate level control system shall maintain constant condensate level in the heating equipment.
- (11) The pH control system shall maintain a constant pH of the decarbonator outlet water.
- (12) The desuperheater or pressure-reducing valve shall be installed if necessary for the heating steam line.
- (13) The instrument air for the plant shall be led from piping of yard instrument air system.

3.11 PLATFORM

The desalination plant shall be provided with ladders and access platform as required for operation and maintenance.

3.12 INSULATION

- (1) Evaporator, brine heater, air ejector with condenser, steam and condensate piping shall be insulated.
- (2) The insulation shall be in conformity with M-003 of "Insulation for Piping and Equipment".

3.13 PAINTING

The painting shall be in conformity with S-420 of "Prime Coat and Shop Finish painting".

3.14 CABLES

- (1) The Bidder shall furnish the power cables, control cables, cable ducts, trays, conduits and grounding wires for the desalination plant.
- (2) Further information regarding Scope of Works and material supply for all items in (1), shall be in accordance with E223 of "Cable and Wires".

3.15 SPECIAL TOOLS AND STANDARDS TOOL SETS

The Bidder shall provide the special tools for the desalination plant.

(Last page of M 440)